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FOUNDATIONS OF AGENTIC AI

LearnPath AI

A multi-agent system for personalized learning and knowledge-gap remediation

Proof-of-concept case study report

Submitted by

Blessings Phiri

PRN: 25070126520

Division: B

Department of AI/ML

SIT, Pune

1. Problem statement and domain justification

Learning builds on earlier learning. A learner who struggles with backpropagation is usually weak in something underneath it, such as the chain rule, gradient descent or matrix shapes, but a quiz score only says "weak in backpropagation". Courses move at one pace for everyone, tutors cannot diagnose every learner, and learners rarely know which gap to fix first or how long it will really take.

Problem statement. Given a learner's goal and time budget, find which knowledge gaps matter and why, plan realistic remediation within the time available, teach and assess with feedback, and adapt when the learner does not improve, while also making sure the study plan and the practice questions are correct before the learner sees them.

Why personalized learning suits agents

- **It is a loop:** diagnose, plan, teach, assess and adapt, not a single answer.
- **It needs multi-step reasoning:** topics form a prerequisite graph, so finding the root cause of a gap means walking that graph.
- **It is verifiable:** plans and quizzes can be checked against the data, which makes a Critic agent useful.
- **It needs humans:** the learner's answers are a human-in-the-loop step, and a human tutor is the fallback when the learner is stuck.

This matches the Education examples in the brief: student performance analysis and personalised academic support.

2. Existing challenge and proposed solution

Today	**LearnPath AI**
One syllabus and pace for everyone	A study path built from the learner's own gaps and time budget
A quiz score per topic ("weak in backpropagation")	A mastery map plus the prerequisite graph: root-cause gaps versus downstream gaps
The learner guesses what to study first	Prerequisites first, weakest first, with hours estimated for each topic
Plans ignore how much time there really is	Honest feasibility: "you need 6.5 h, you have 4 h, extend to 7 days"
Failing a quiz means repeating the same material	Re-teach with a new explanation, easier resources and the missed question; two rounds, then a human tutor
Generated study content is trusted blindly	A Critic validates every plan and quiz against the source data

The benefits are design goals, not measured results. This report evaluates whether the workflow behaves as designed on dummy data, not how much real learners would gain.

Why not a plain chatbot? A chatbot would answer one question at a time from the model's memory. This problem needs private learner data, reasoning over a prerequisite graph, time-budget calculations, an assessment loop that changes the plan, and independent validation. Those are separate responsibilities, so they are separate agents.

3. Agent roles, responsibilities and orchestration

Agent	Responsibility
Planner	Parses the goal (topics, days, hours per day) with rules or an LLM, applies privacy and academic-integrity guardrails, routes the request (refuse, escalate or learn) and writes the task plan.
Diagnostic (research)	Calls tools for the learner profile, mastery map and prerequisite closure. Finds root-cause gaps and downstream gaps and estimates the hours each topic needs.
Tutor (executor)	Builds the day-by-day study path within the time budget (deferring or partly covering what does not fit), writes explanations, prepares quizzes and re-teaches after a failed quiz.
Critic / Assessor	Re-derives the facts from raw data to validate each plan and quiz. Grades the learner's answers, updates mastery and decides: cleared, re-teach or escalate.
Orchestrator	LangGraph state machine with a human-in-the-loop pause for the learner's answers, retry loops and escalation tickets.

Orchestration workflow

1. **Plan.** The Planner parses the goal and routes it. It refuses off-topic requests, other learners' data and requests to do assignments; requests for a human go to a tutor ticket.
2. **Diagnose.** The Diagnostic agent builds the mastery map and the root-cause analysis.
3. **Build and validate the plan.** The Tutor builds the study path and the Critic checks it with 8 checks. If a student fails, the Tutor rebuilds the study plan with the feedback (maximum 2 attempts), then the case is escalated.
4. **Teach and quiz.** For each planned topic the Tutor explains and prepares a quiz, and the Critic validates the quiz with 7 checks.
5. **Pause for the learner.** The graph waits for the learner's answers (or a simulated learner in autopilot mode).
6. **Assess and decide.** The Assessor grades and updates mastery. Cleared: next topic. Not cleared: re-teach. Not cleared after 2 rounds: escalate to a human tutor and move on.
7. **Report.** A progress report shows before and after mastery, the study path, practice results and next steps.

Key rules

- **Mastery** is recency-weighted accuracy over past quizzes. Mastered at 70% or more, shaky at 50 to 69%, a gap below 50%. Topics with no history are assumed at 40% and flagged as unassessed.

- **Hours needed** = estimated hours x (1 - mastery), rounded to the nearest half hour, with a minimum of half an hour.
- **Root cause** = a topic that needs work and whose prerequisites do not. Every gap blocked behind it is downstream.
- **Over budget:** schedule in prerequisite order until the time runs out, partly cover one topic, defer the rest, and state how many days are really needed.
- **Cleared** = at least 2 of 3 questions correct and updated mastery of at least 70% (after studying, the latest quiz counts double).
- **No gaps:** challenge mode with harder questions and a stretch topic that the goal unlocks.

4. Architecture

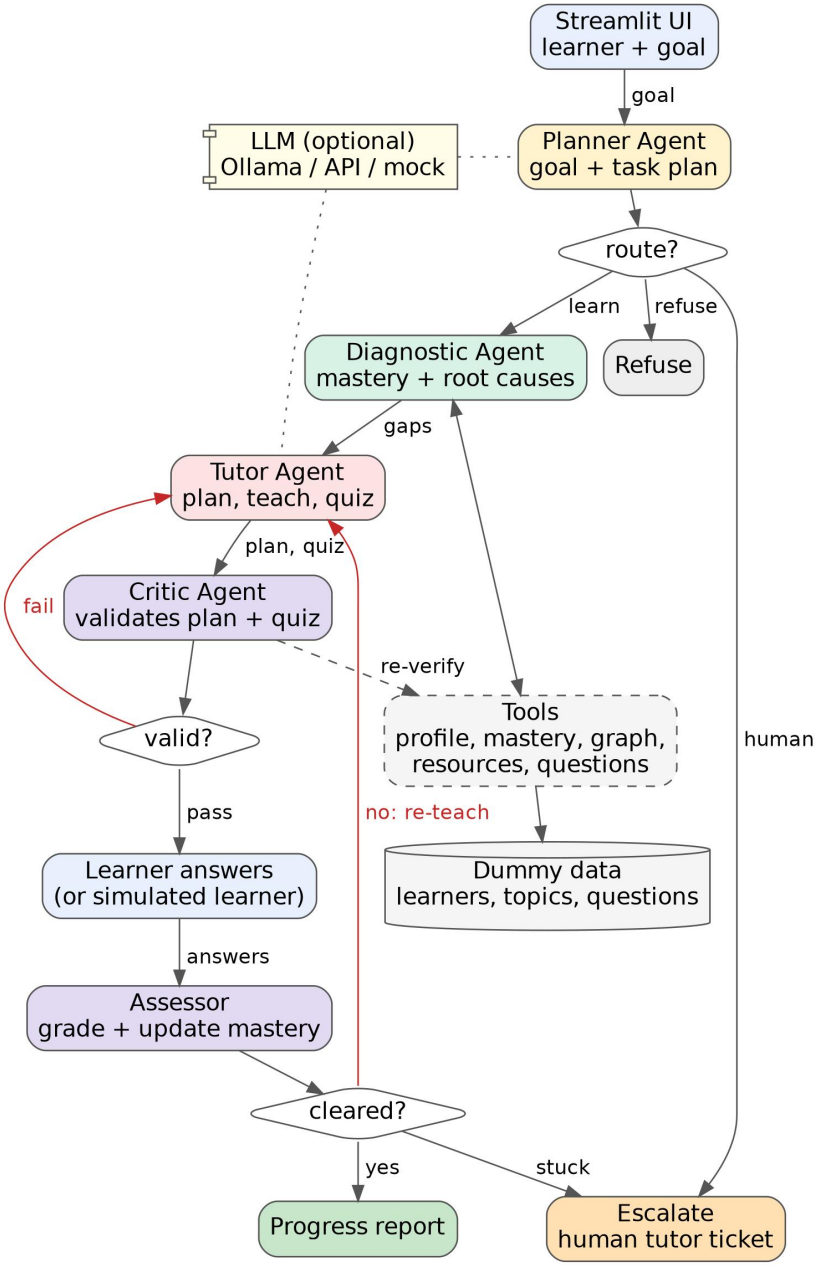


Figure 1. LearnPath AI architecture: agents, tools, data flow and decision points

Data flow: the goal becomes a structured goal and a task plan. The Diagnostic agent turns tool results (mastery map, prerequisite closure) into a diagnosis. The Tutor turns the diagnosis into a study plan and quizzes. The Critic checks each plan and quiz and either approves it or sends feedback back. Approved quizzes go to the learner, whose answers are graded by the Assessor, which decides whether to move on, re-teach or escalate. The Critic re-queries the source data instead of trusting the Tutor, so a wrong number, an impossible schedule or a bad answer key cannot validate itself.

5. Proof of concept

Technology

Python, Streamlit (UI), LangGraph (orchestration, with an interrupt for the learner's answers), LangChain tools, Pydantic. Models are optional: the built-in mock mode needs no key and no internet, Ollama provides free local models, and an API model can be selected in the sidebar.

Tools

Tool	Used by	Returns
get_learner_profile	Diagnostic	Name, level, amount of quiz history
compute_mastery_map	Diagnostic, Critic	Mastery and status for every topic
get_prerequisite_closure	Diagnostic	Goal topics plus all prerequisites, in learning order
find_resources	Tutor	Videos, articles and exercises for a topic and level
get_practice_questions	Tutor	Questions matching the wanted difficulty, avoiding seen ones
escalate_to_tutor	Orchestrator	Opens a human tutor ticket

Dummy data

The data folder holds 12 machine-learning topics in a prerequisite graph, 60 multiple-choice questions (5 per topic, difficulty 1 to 3), 48 study resources and 4 learners. The learners are designed to trigger different behaviour.

ID	Learner	Goal	Designed to show
L001	Ananya Rao	Backpropagation, 10 days x 1.5 h	Root causes (calculus, linear algebra) behind a hard topic; Gradient Descent needs a re-teach
L002	Rahul Mehta	Model evaluation and regularization, 5 days x 1 h	One root cause (evaluation) blocking another topic
L003	Alice Daniels	Linear regression, 4 days x 1 h	Not enough time (needs 6.5 h, has 4 h); a stuck topic escalated to a tutor
L004	Karan Shah	Neural networks, 3 days x 1 h	No gaps: challenge mode, a second round and a stretch topic

Optional LLM (Ollama or an API)

The system works with no LLM. When one is selected it is used in four places, and every use has a fallback. Privacy and integrity guardrails are decided by rules, never by the LLM.

Agent	What the LLM may do	If it fails or is off
Planner	Parse a free-text goal into topics and a time budget	Rule-based parser
Tutor	Write the explanation for each topic and re-teach round	Template built from the topic notes
Tutor	Write extra practice questions (off by default)	Curated question bank
Critic	Answer an LLM-written question independently to verify its answer key	The question is rejected

Every LLM call is counted and shown in the agent trace. LLM explanations are guard-checked for length and relevance. A small local model that cannot produce structured output simply causes a fallback to the rules.

User interface

The sidebar selects the learner, the model mode, what the LLM may do, the practice mode (answer the questions yourself, or autopilot with a simulated learner) and a switch that injects faulty first drafts so the Critic can be demonstrated. The tabs are: Learning path, Knowledge gaps (the prerequisite graph coloured by mastery, at the start and now), Practice (study material, quiz form and instant feedback), Report, Agent trace, Critic validation, Architecture and Evaluation.

How to run

1. Install Python 3.10 or newer and open the project folder in VS Code or a terminal.
2. Create and select a virtual environment, then install the packages: `pip install -r requirements.txt`
3. Start the app: `streamlit run app.py` and open http://localhost:8501.
4. Run the tests without the UI: `python -m learnpath.evaluation`
5. Optional, for Ollama: install Ollama, run `ollama pull llama3.1`, run `pip install langchain-ollama`, then choose ollama in the sidebar and press Check Ollama connection.

6. Output validation

The Critic does not trust the Tutor's working. It re-derives mastery and the required work from the raw data and checks every plan and quiz against it.

Plan checks

Check	What it catches
Total hours	A summary that does not match an independent recomputation
Prerequisite order	A topic scheduled before a prerequisite it depends on
Time budget	Total or daily time above what the learner has
No wasted time	Time spent on topics the learner has already mastered
Root-cause coverage	A root-cause gap that is neither scheduled nor openly deferred
Resources	Resources that do not exist, are about another topic, or are too advanced for a weak learner
Honest feasibility	A plan that hides a time shortfall
Hours per topic	Allocated time that does not match the diagnosis

Quiz checks

Check	What it catches
Format	Empty text, duplicate options, or an answer key outside the options
Topic	Questions about a different topic
Count and difficulty	The wrong number of questions or questions harder than requested
Duplicates	The same question twice
Explanations	Questions without feedback for the learner
Freshness	Re-asking a question the learner already answered correctly when fresh ones exist
LLM self-check	An LLM-written question whose answer key the model cannot reproduce

7. Results

The tests below run in mock mode with a simulated learner, using python -m learnpath.evaluation or the Evaluation tab of the app.

Test	Outcome
11 scenarios: 5 learning paths, 4 refusals (another learner's data, off-topic, an assignment request, an unknown topic), 1 request for a human tutor, 1 early stop	11/11 passed
Fault injection: a corrupted first study plan and a corrupted first quiz in each of the 5 learning-path scenarios	5/5 rejected by the Critic and recovered, with the same outcome as the clean run
Question picker sweep: every topic, difficulty pattern and combination of missed questions	288/288 quizzes pass the Critic

Example learner journeys (simulated learner)

Learner	What happened
L001 Ananya	Root causes found: Calculus and Linear Algebra. Plan: Calculus, Linear Algebra, Linear Regression, Gradient Descent, Neural Networks, Backpropagation (10.5 h of 15 h). Gradient Descent failed the first quiz (0/3), was re-taught and cleared. Mastery: Calculus 30% to 77%, Linear Algebra 60% to 87%, Gradient Descent 30% to 77%, Backpropagation 40% (assumed) to 100%.
L002 Rahul	One root cause, Model Evaluation, blocking Regularization. 3 h of 5 h needed. Both cleared: 40% to 80% and 30% to 77%.
L003 Alice	Needs 6.5 h, has 4 h. The plan says so, recommends 7 days, covers Calculus fully and Linear Algebra partly, and defers Linear Regression. Calculus failed two rounds (0/3 each) and was escalated to a human tutor.
L004 Karan	No gaps. Challenge mode on Neural Networks (70%): the first challenge quiz was not enough, a second round cleared it (to 90%), then the stretch topic Backpropagation was cleared.

Worked example of the Critic

With the faulty-draft switch on, the first study plan for L001 is corrupted: the last topic is moved to the front, a topic the learner has already mastered is inserted, the total hours are inflated by 1.5 and one resource id does not exist. The Critic fails 6 of the 8 plan checks and sends the findings to the Tutor, which rebuilds a valid plan. The first quiz is also corrupted (an invalid answer key and a question from another topic); 2 quiz checks fail and a valid quiz is prepared.

These are functional tests on dummy data with a simulated learner. They show that the workflow, routing and validation behave as designed. They do not measure learning gains with real learners.

8. Limitations

- The data is dummy and the evaluation uses a simulated learner. No real learning gains were measured.
- Mock mode uses keyword rules for goals and templates for explanations. Real LLM output using an Ollama model should be exercised instead for better explanations.
- Mastery is a simple estimate from a few quizzes, and the question bank is small (5 questions per topic).
- The Critic catches wrong plans and malformed questions, not weak teaching. The correctness of an LLM-written question rests on a self-consistency check, not on ground truth.
- The learner is chosen in the UI. A real system needs authentication.

9. Future scope

- Real course content, a larger question bank, flash cards, and spaced-repetition review of cleared topics.

- Retrieval over lecture notes (RAG) for explanations, and MCP servers for the learning management system and the calendar.
- Better mastery models such as item response theory or Bayesian knowledge tracing, and open-ended answers graded with a rubric.
- A pilot with real learners, LangSmith tracing, and an LLM judge alongside the rule-based Critic.

10. Conclusion

LearnPath AI shows an agentic workflow rather than a chatbot: a Planner breaks down the goal, a Diagnostic agent finds root causes in a prerequisite graph, a Tutor plans within the time budget and teaches, and a Critic and Assessor validate plans and quizzes, grade answers and decide whether to move on, re-teach or escalate. A human-in-the-loop pause, bounded retry loops, honest feasibility reporting and a human tutor fallback make it adaptive and safe. On dummy data it passed every scenario in the test suite and caught every deliberately injected error, and it can use a free local model such as Ollama without depending on one.